

Defendable Rules for LLM Rationale Evaluation in Banking Governance: A Multi-Source Provenance Framework

Javier Marín
Independent AI Consultant
Groundlens
javier@groundlens.dev

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Abstract

Banking governance increasingly relies on Large Language Models (LLMs) for credit decisioning, anti-money-laundering (AML) review, Know-Your-Customer (KYC) verification, and adjacent rationale-driven workflows. LLMs are fallible: confidently-wrong outputs — “hallucinations” — create direct regulatory and operational risk in these settings. Rule-based evaluation layers that triage rationale quality are emerging as a deployment requirement under the revised US model-risk guidance SR 26-2 ([Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026](#)), the EU AI Act ([European Parliament and Council, 2024](#)), the Bank of England’s SS1/23 ([Bank of England, Prudential Regulation Authority, 2023](#)), and the NIST AI Risk Management Framework ([National Institute of Standards and Technology, 2023, 2024](#)). However, existing approaches to rule-based rationale evaluation in this setting either derive their criteria from single research lineages with limited cross-source triangulation, or articulate principles at the system-governance level rather than at the individually-evaluable rationale level. Neither path produces a rule set whose individual rules carry verifiable academic, industrial, and regulatory provenance. We present a multi-source provenance framework that triangulates rationale-evaluation rules across five independent research tracks: peer-reviewed Natural Language Processing (NLP) literature on hallucination evaluation, tier-1 bank and asset-manager public reports, banking-regulator whitepapers, cross-industry AI governance frameworks, and financial-domain NLP benchmarks. From an initial pool of 110 candidate criteria, we identify 20 canonical rules whose provenance is verifiable across at least three independent sources, organized into five empirically-emergent sub-score categories (*Groundedness, Completeness, Calibration, Traceability, Robustness*). The resulting rule set is implemented as the open-source `groundlens_banking_rules` reference, designed to be defendable in front of EU and US banking supervisors and Big Four model-risk auditors.

Keywords: LLM evaluation, hallucination detection, banking AI governance, SR 26-2, EU AI Act, model risk management, retrieval-augmented generation, rationale faithfulness.

1 Introduction

The deployment of Large Language Models in retail and corporate banking is no longer hypothetical. Morgan Stanley reports an internal LLM-based “AI @ Morgan Stanley Assistant” built jointly with OpenAI, gated by an explicit evaluation framework that subjects every use case to regression testing prior to release ([Morgan Stanley and OpenAI, 2024](#)). Bank of America’s “Erica” assistant exposes over 50 million users to LLM-assisted customer-facing rationales ([Bank of America, 2024](#)). HSBC publishes an explicit set of principles for the ethical use of AI ([HSBC](#)

Holdings, 2024), requiring that any AI-driven decision be justifiable as if it had been made by a human under the same standards. UBS publicly acknowledges that LLMs can be “simultaneously highly confident and wrong” as a design input (UBS Group, 2024), and the Monetary Authority of Singapore’s MindForge consortium names hallucination, prompt injection, and data leakage as first-class risk dimensions for generative AI in banking (Monetary Authority of Singapore, 2024).

In parallel, banking supervisors have published increasingly specific evaluation requirements for AI systems. In the United States, the Board of Governors of the Federal Reserve System, the Federal Deposit Insurance Corporation, and the Office of the Comptroller of the Currency jointly issued SR 26-2 (Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026), which supersedes both SR 11-7 (April 2011) and SR 21-8 (April 2021) and establishes the revised model-risk-management standard against which any model used in a US bank — including LLM-based ones — is supervised. The Bank of England’s PRA SS1/23 (Bank of England, Prudential Regulation Authority, 2023) provides the UK analogue. The European Central Bank’s revised Guide to Internal Models (European Central Bank, 2024) addresses Machine Learning (ML) explainability for credit Internal Ratings-Based (IRB) applications. The European Union has enacted the AI Act (European Parliament and Council, 2024), whose Articles 12–15 directly constrain the design and documentation of high-risk AI systems used in credit scoring, employment, and adjacent banking activities. The NIST AI RMF 1.0 (National Institute of Standards and Technology, 2023) and its Generative AI Profile NIST AI 600-1 (National Institute of Standards and Technology, 2024) provide the dominant US voluntary referent, with the latter explicitly naming “confabulation” (hallucination) as a measurable risk category.

In this environment, banks deploying LLM-assisted rationale generation in credit, AML, KYC, sanctions, fraud, concentration, and model-risk workflows face a concrete operational question: *against what set of criteria should LLM rationales be evaluated, and how can the chosen criteria be defended to a regulator or auditor?*

Existing proposals for rule-based rationale evaluation in this setting fall into two broad categories. First, single-source checklists derived from a specific research lineage: these are internally coherent but lack the cross-source independence that supervisors look for when evaluating model-validation evidence under SR 26-2 (Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026) or analogous EU and UK guidance. Second, system-level governance frameworks — the FEAT principles (Monetary Authority of Singapore, 2018), the NIST AI Risk Management Framework (National Institute of Standards and Technology, 2023), ISO/IEC 42001 (International Organization for Standardization, 2023b) — which articulate evaluation vocabulary at the right level of abstraction for organizational governance but do not provide pattern-matchable rules applicable to individual rationales. The present paper occupies the gap between these two categories: a rule-by-rule rule set whose individual criteria are triangulated across multiple independent source types.

Contribution. This paper presents a methodology — and its execution — for constructing a rule set for LLM rationale evaluation in banking governance whose provenance can be triangulated across multiple independent source categories. Concretely, we make four contributions:

1. A *five-track parallel research design* (peer-reviewed NLP, tier-1 bank reports, banking regulator whitepapers, cross-industry frameworks, financial NLP benchmarks) for source-independent evidence on rationale-evaluation criteria (§3).
2. A *co-occurrence ranking procedure* that selects rules based on the number of independent source tracks supporting each criterion, with deduplication and semantic merging across tracks (§3.2).

3. An *empirically-derived skeleton* of five sub-score categories (Groundedness, Completeness, Calibration, Traceability, Robustness) that emerges from clustering the top-ranked rules, and that aligns with regulator vocabulary rather than imposing an ex-ante taxonomy (§4.1).
4. A *canonical top-20 rule set* (§4.2), each rule documented with at least three independent provenance citations following the ideal pattern of one academic, one industry, and one regulator source, together with 30 honorable mentions reserved for domain-specific extensions (§4.3).

Scope. The rule set is designed to be *cross-domain* within banking governance — applicable to credit decisioning, AML, KYC, fraud, sanctions, concentration, and model-risk rationales — rather than tuned to a single sub-domain. Domain-specific extensions (AML SAR-narrative rules, credit adverse-action reason codes) are intentionally kept in the honorable-mentions tier, available for composition by deployers who need them. The set is implemented and shipped as the open-source `groundlens_banking_rules` reference in the GroundLens library; the implementation is companion work and not the subject of this paper.

2 Related Work

2.1 Hallucination and faithfulness evaluation in NLP

The NLP community’s treatment of LLM hallucination as a first-class evaluation target has crystallized over the past three years. Ji et al. (2023), in the ACM Computing Surveys, frame hallucinations along the dichotomy of *intrinsic* (contradicting source input) versus *extrinsic* (unverifiable from source) and treat *faithfulness* as distinct from *factualness*. This taxonomy underlies the more operational metrics that follow.

Min et al. (2023) introduce FactScore, which decomposes a generation into atomic facts and computes the fraction supported by a reliable knowledge source; the motivation, quoted from their abstract, is that “generations often contain a mixture of supported and unsupported pieces of information, making binary judgments of quality inadequate.” Es et al. (2024) (RAGAs) generalize this idea to retrieval-augmented generation, defining *Faithfulness*, *Answer Relevance*, and *Context Relevance* as three orthogonal metrics. RAGAs Faithfulness is operationalized as a sentence-level statement-extraction procedure followed by per-statement verification against the retrieved context, with score $F = |V|/|S|$ where $|V|$ is the number of statements supported.

Manakul et al. (2023) (SelfCheckGPT) take a complementary route: rather than verifying against an external knowledge source, they sample multiple generations and treat *divergence across samples* as evidence of hallucination, on the rationale that “if an LLM has knowledge of a given concept, sampled responses are likely to be similar and contain consistent facts. However, for hallucinated facts, stochastically sampled responses are likely to diverge and contradict one another.” Liu et al. (2023) (G-Eval) use the LLM itself as judge, with chain-of-thought-generated evaluation steps and form-filling output, and report a self-preference bias in which LLM judges over-rate LLM-generated text. Li et al. (2023) construct HaluEval, a large-scale hallucination benchmark with explicit type taxonomy, and find that supplying “external knowledge or adding reasoning steps can help LLMs recognize hallucinations.”

For RAG specifically, Saad-Falcon et al. (2024) present ARES, factoring evaluation into Context Relevance, Answer Faithfulness, and Answer Relevance, which has since been adopted as the de facto triad and is mirrored in the TruLens framework (TruLens Contributors, 2024). Chen et al. (2024) (RGB) extend RAG evaluation into four *abilities*: noise robustness, negative rejection, information integration, and counterfactual robustness. Friel et al. (2024) (RAGBench) propose the *TRACe* framework: utilization, relevance, adherence, completeness. Su et al. (2024) address cross-source conflict in retrieved contexts as a distinct evaluation axis.

For free-text rationale evaluation specifically, [Camburu et al. \(2018\)](#) (e-SNLI) introduce natural-language explanations of model decisions as themselves an evaluable artifact, and [Chen et al. \(2023\)](#) (REV) operationalize the question of whether a rationale contributes *new, label-relevant information* beyond the input or the label via conditional V-information.

For factual consistency at the document level, [Honovich et al. \(2022\)](#) (TRUE) re-evaluate factuality metrics on a unified benchmark and find that large-scale NLI and question-generation-and-answering (QAG) approaches achieve strong, complementary results. [Lin et al. \(2022\)](#) (TruthfulQA) measure the propensity of models to reproduce common human misconceptions, framing truthfulness as “avoiding generating false answers learned from imitating human texts.”

For the linguistic substrate, three older works are load-bearing. [Grice \(1975\)](#) establishes the cooperative principle of conversation, of which the Maxim of Quantity (“Make your contribution as informative as is required”) is the normative basis for specificity criteria. [Hyland \(1998\)](#) provides the canonical taxonomy of *hedging* in scientific writing, distinguishing epistemic from approximative hedges. [Vincze et al. \(2008\)](#) (BioScope) and [Farkas et al. \(2010\)](#) establish corpus-driven hedge detection in biomedical text, and [Louis and Nenkova \(2011\)](#) train an automatic classifier for sentence-level specificity.

2.2 Banking AI governance frameworks

Banking-sector frameworks for AI governance run in parallel to the academic literature, with limited cross-citation. The earliest substantial sectoral framework is the Monetary Authority of Singapore’s Fairness, Ethics, Accountability, Transparency (FEAT) principles ([Monetary Authority of Singapore, 2018](#)), subsequently operationalized by the Veritas Consortium with four assessment-methodology PDFs ([Monetary Authority of Singapore Veritas Consortium, 2022](#)). FEAT vocabulary appears, with relatively little adaptation, in HSBC’s published principles ([HSBC Holdings, 2024](#)) and in the substantive structure of several Big Four advisory frameworks ([Deloitte, 2024](#); [Ernst & Young, 2024](#); [KPMG International, 2023](#)).

For generative AI specifically, the MAS-led MindForge project ([Monetary Authority of Singapore, 2024](#)) extends FEAT with seven risk dimensions including hallucination, prompt injection, and data leakage, and is the only publicly-available regulator-backed banking framework that names hallucination explicitly. The NIST AI 600-1 Generative AI Profile ([National Institute of Standards and Technology, 2024](#)) provides the US analogue, naming “confabulation” as one of twelve GenAI-unique risk categories.

Among tier-1 banks, public disclosure of evaluation criteria is uneven. The most substantive public document is HSBC’s 9-page principles PDF ([HSBC Holdings, 2024](#)), organized around seven principles that cover purpose legitimacy, justifiability, fairness review, pre- and post-deployment monitoring, third-party parity, accountability, and individual rights. JP Morgan’s public position is articulated as the integration of AI use cases into Model Risk Governance ([JPMorgan Chase, 2023](#)) but is not accompanied by a published rule set. Morgan Stanley’s collaboration with OpenAI ([Morgan Stanley and OpenAI, 2024](#)) is the most operationally specific public source, describing an evaluation framework that subjects every use case to documented eval and daily regression testing. UBS ([UBS Group, 2024](#)), Citi ([Citi Ventures, 2024](#)), and Bank of America ([Bank of America, 2024](#)) publish governance principles in public statements but no formal published frameworks. BlackRock, Goldman Sachs, Deutsche Bank, Barclays, and Santander were searched: none publish detailed AI rationale evaluation frameworks publicly as of the present writing.

2.3 Banking regulator whitepapers on AI/ML and GenAI

The US model-risk lineage is currently anchored by SR 26-2 ([Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026](#)), the revised interagency guidance issued jointly by the Federal Reserve, the

FDIC, and the OCC in April 2026. SR 26-2 supersedes both the original SR 11-7 (2011) and SR 21-8 (the 2021 BSA/AML interagency statement), and reflects fifteen years of supervisory experience and significant advancements in modeling practices, including the widespread adoption of machine-learning techniques. The OCC Comptroller’s Handbook on Model Risk Management ([Office of the Comptroller of the Currency, 2021](#)) provides operational interpretation. In the UK, the PRA’s SS1/23 ([Bank of England, Prudential Regulation Authority, 2023](#)) states the model-risk-management principles with adaptation for ML, and the joint Bank of England / PRA / FCA Discussion Paper DP5/22 ([Bank of England and Financial Conduct Authority, 2022](#)) addresses AI/ML in financial services with the UK government’s five cross-cutting principles (safety, transparency and explainability, fairness, accountability and governance, contestability and redress).

In the EU, the ECB’s revised Guide to Internal Models ([European Central Bank, 2024](#)) addresses ML explainability for credit IRB applications. The European Banking Authority’s loan-origination and monitoring guidelines ([European Banking Authority, 2020](#)) require traceability and auditability of underlying models, and the EBA Guidelines on PD estimation, LGD estimation and the treatment of defaulted exposures ([European Banking Authority, 2017](#)) require institutions to consult business experts on the rationale and risk contribution of selected risk drivers. The EU AI Act ([European Parliament and Council, 2024](#)) is the binding cross-sector regulation; its Articles 12 (Record-Keeping), 13 (Transparency), 14 (Human Oversight), 15 (Accuracy, Robustness, Cybersecurity), and 17 (Quality Management) apply directly to high-risk AI systems used in credit scoring, employment, and adjacent banking activities. ([Bank for International Settlements, Financial Stability Institute, 2024](#)) (BIS FSI Insights No. 63) provides a global supervisory perspective and explicitly names NIST AI RMF as the international benchmark.

2.4 Cross-industry frameworks and standards

The NIST AI RMF 1.0 ([National Institute of Standards and Technology, 2023](#)) and the GenAI Profile NIST AI 600-1 ([National Institute of Standards and Technology, 2024](#)) dominate the US voluntary-framework landscape. ISO/IEC 42001 ([International Organization for Standardization, 2023b](#)) provides an AI Management Systems standard analogous to ISO 27001 for information security; ISO/IEC 23894 ([International Organization for Standardization, 2023a](#)) addresses AI risk management. The Singapore AI Verify framework ([Infocomm Media Development Authority \(Singapore\), 2024](#)) provides a testing toolkit. The OECD AI Principles ([Organisation for Economic Co-operation and Development, 2019](#)) and the EU AI Act high-risk Annex form the principal international policy referents.

2.5 Financial-domain NLP benchmarks

For numerical reasoning over financial reports, [Chen et al. \(2021\)](#) introduce FinQA with explicit *execution accuracy* and *program accuracy* metrics — the answer must be both numerically correct and produced by a logically correct reasoning program. [Islam et al. \(2023\)](#) (FinanceBench) frame evidence-string grounding as a core requirement, defining for each entry “an evidence string (which contains the information needed to verify that the answer to the question is correct) and a page number from the relevant document,” and report a refusal-rate analysis demonstrating that “GPT-4-Turbo used with a retrieval system incorrectly answered or refused to answer 81% of questions.” Domain-specific LLM evaluation for AML and credit decisioning is emerging ([Anonymous authors, 2025](#); [Krivorotov and Richey, 2022](#)), with stability of post-hoc explanations against adverse-action requirements as a recurrent theme.

2.6 The provenance gap

What is missing across this corpus, and what motivates the present paper, is a *rule-by-rule cross-source provenance audit* of the criteria that should constitute a canonical rule set for LLM rationale evaluation in banking governance. Existing single-source checklist proposals — of which [De La Chica and Martí-González \(2026\)](#) is a recent example — do not triangulate individual rules across academic, industry, and regulator sources. RAGAs, ARES, and the TRACe frameworks define metrics but are not contextualized to banking governance vocabulary. Regulator whitepapers articulate principles but stop short of pattern-matchable rules. Tier-1 bank disclosures provide vocabulary but are intentionally non-operational. The contribution of this paper is to perform that triangulation explicitly.

3 Methodology

3.1 Five-track parallel research design

A central design choice in the construction of the rule set is the partitioning of source candidates into five independent tracks before any ranking or merging takes place. The motivation is statistical: a rule that appears in three sources from the same research lineage (*e.g.*, three NLP papers that cite each other) provides less independent evidence than a rule that appears once in an NLP paper, once in a banking regulator whitepaper, and once in a tier-1 bank disclosure. Independence is the relevant axis for regulator-defendable provenance.

The five tracks are:

1. **Track 1 — Peer-reviewed NLP literature** on hallucination evaluation, rationale faithfulness, RAG benchmarks, and the linguistic substrate (Grice, Hyland, BioScope, CoNLL 2010, Louis & Nenkova). Source priority: EMNLP, ACL, NAACL, EACL, NeurIPS, ACM Computing Surveys.
2. **Track 2 — Tier-1 bank and asset-manager public reports** on AI principles, governance frameworks, and operational evaluation practices, scoped to BlackRock, JPMorgan, Goldman Sachs, Morgan Stanley, Barclays, UBS, Deutsche Bank, Citi, Bank of America, HSBC, Santander, plus the MAS Veritas industry consortium.
3. **Track 3 — Banking regulator whitepapers and supervisory statements** from the Federal Reserve, OCC, FDIC, ECB, Bank of England, FCA, EBA, ESMA, BIS, Banco de España, BaFin, ACPR, plus the NIST AI RMF family and the EU AI Act.
4. **Track 4 — Cross-industry frameworks, consortia, and Big Four advisory reports:** MAS FEAT/Veritas, NIST AI RMF, OECD AI Principles, ISO/IEC 42001 and 23894, Singapore AI Verify, AFME, ISDA, FinRegLab, PwC, Deloitte, KPMG, EY.
5. **Track 5 — Financial-domain NLP benchmarks and RAG evaluation tooling:** FinQA, FinanceBench, ARES, RGB, RAGBench, ConflictBank, CRUD-RAG, TruLens, Vectara HHEM, BIG-bench HHH, and the emerging AML/credit-XAI literature.

Each track was searched independently using a dedicated retrieval and extraction agent. Each agent was instructed to: (a) identify the most authoritative sources for its track; (b) fetch the primary documents (PDFs, papers, framework specifications); (c) extract pattern-matchable evaluation criteria with verbatim quotation; (d) record provenance (source, year, section, URL) for every criterion; (e) flag failed fetches honestly rather than substituting paraphrase from memory.

3.2 Co-occurrence ranking procedure

After parallel extraction, the corpus contained 110 candidate criteria distributed unevenly across tracks. We deduplicated semantically: two candidates were merged if they expressed the same constructive criterion regardless of vocabulary — for example, RAGAs *Faithfulness*, TRACe *Adherence*, ARES *Answer Faithfulness*, and NIST 600-1 *Confabulation* all express the same constructive criterion (*claims supported by context*) and were merged into a single canonical rule.

For each canonical rule r , we computed a *cross-track support score*:

$$S(r) = \sum_{t=1}^5 \mathbb{I}[r \text{ appears in track } t], \quad (1)$$

where each track contributes at most one unit of support regardless of how many sources within that track corroborate the rule. This formulation deliberately under-weights within-track redundancy and rewards cross-track independence.

Rules were then ranked by $S(r)$ in descending order. Rules with $S(r) \geq 3$ are considered to have triangulated provenance and are candidates for the top-20 rule set; rules with $S(r) \leq 2$ are retained as honorable mentions for future expansion. Where multiple rules share the same $S(r)$, ties are broken by: (i) presence of a regulator citation, (ii) presence of a tier-1 bank citation, and (iii) breadth of cross-domain applicability within banking governance.

3.3 Deduplication and merging procedure

Two criteria c_1 and c_2 from different tracks are merged into a single canonical rule when:

1. Both express the same operational test on a rationale ($c_1 \equiv c_2$ as functions of (question, response, context)).
2. Their vocabulary differs but the implementation logic would be substantively identical.
3. Neither adds an axis of evaluation absent from the other.

We are explicit about merge decisions in §4.2: each top-20 rule lists its constituent merged sources, which is the basis of the cross-track support score.

3.4 Skeleton derivation by category clustering

Rather than impose an ex-ante taxonomy on the rule set, we derived the sub-score categories empirically by clustering the 20 ranked rules according to the operational dimension each measures. Five categories emerged: *Groundedness* (claim-to-source linkage), *Completeness* (coverage of the question’s required dimensions), *Calibration* (appropriate expression of uncertainty), *Traceability* (citation, audit, and validation references), and *Robustness* (resistance to noisy or contradictory retrieval). The categories align closely with the converging vocabulary of regulator and cross-industry frameworks; we discuss this alignment in §5.1.

3.5 Source verification

Every quotation extracted from a primary source was verified by direct fetch of the source document — arXiv PDF, the EU AI Act official-text mirror at artificialintelligenceact.eu, or the regulator’s published PDF. Where a quoted phrase that initially appeared in an authoritative summary could not be located in the primary source, the citation was either replaced with a verbatim phrase that actually appears in the primary source, or rewritten as a paraphrase that does not put words in the regulator’s mouth. The deduplication and merging decisions in §3.3, the skeleton clustering in §3.4, and the weight assignments in §4.2 were performed by a single rater (the author); inter-rater reproducibility remains as future work.

A schematic of the methodology is shown in Figure 1.

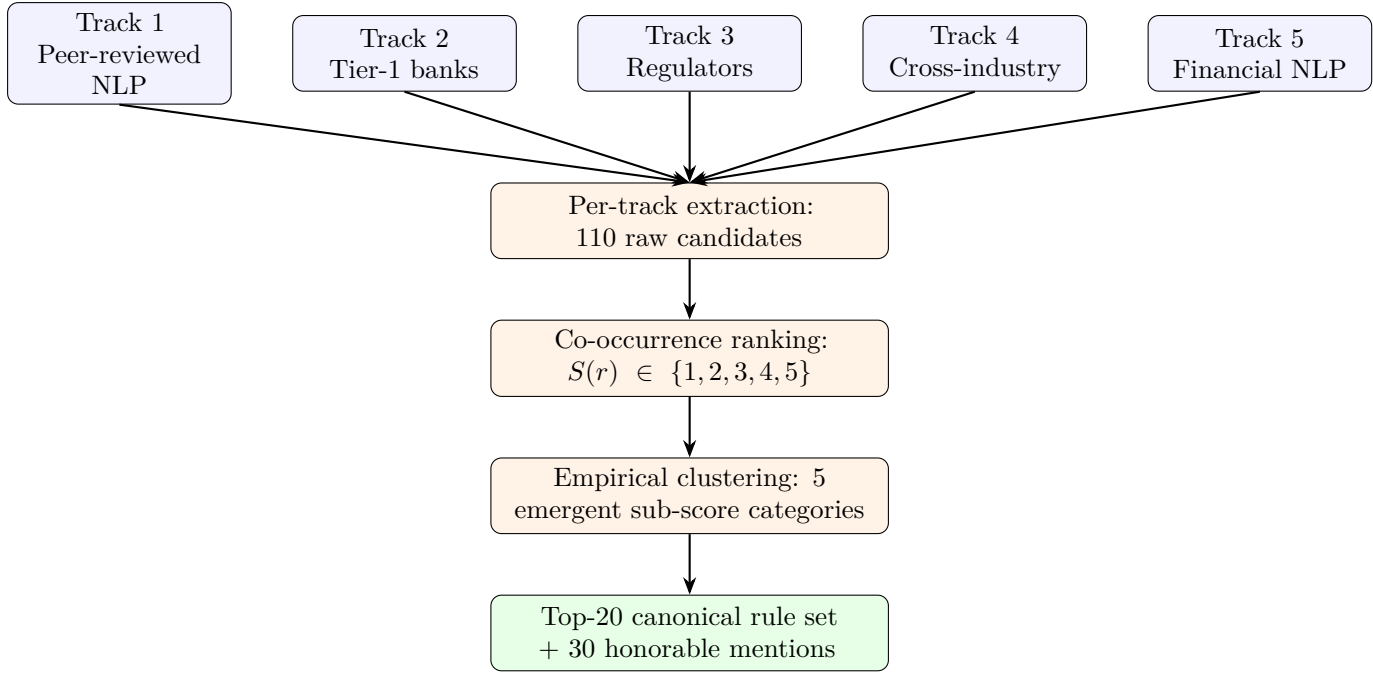


Figure 1: Five-track parallel research design. Each track contributes at most one unit of support per rule, so cross-track presence is the dominant signal in the ranking.

4 Results

4.1 Emergent skeleton: five sub-score categories

The empirical clustering of the top-20 rules yielded five sub-score categories, summarized in Table 1. Two categories — Groundedness and Calibration — received support from all five research tracks; Completeness and Traceability received support from four tracks; Robustness received support from three tracks but appeared in explicit form in NIST AI 600-1 (National Institute of Standards and Technology, 2024), EU AI Act Art. 15 (European Parliament and Council, 2024), and RGB (Chen et al., 2024), providing sufficient cross-source independence.

Sub-score	Rules	Strongest cross-track signal
Groundedness	5	All five tracks
Completeness	3	Tracks 1, 3, 4, 5
Calibration	4	All five tracks
Traceability	5	All five tracks
Robustness	3	Tracks 3, 4, 5
Total	20	

Table 1: Emergent skeleton derived from clustering the top-20 rules. Cross-track signal counts the number of independent research tracks contributing at least one supporting citation to any rule in the category.

The five-category skeleton aligns closely with the converging vocabulary identified in Track 4: fairness, transparency, accountability, robustness, and reliability appear in 13 or 14 of the 14 frameworks surveyed, with traceability close behind at 12 of 14 (Deloitte, 2024; Ernst & Young, 2024; European Parliament and Council, 2024; Infocomm Media Development Authority (Singapore), 2024; International Organization for Standardization, 2023a,b; KPMG International, 2023; Monetary Authority of Singapore, 2018; Monetary Authority of Singapore Veritas Consor-

tium, 2022; National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, 2019). Groundedness and Calibration are the two LLM-era additions that emerge specifically from the post-2023 GenAI risk literature (FinRegLab, 2023; Monetary Authority of Singapore, 2024; National Institute of Standards and Technology, 2024). Robustness recurs across regulators and RAG benchmarks but is sometimes folded into other categories in framework documents.

4.2 Top 20 rules with provenance

Each of the 20 rules is presented below with: (i) a stable identifier; (ii) the sub-score category and proposed weight; (iii) a plain-English description; (iv) the cross-track support count; (v) full provenance with verbatim quotation from at least three independent sources; (vi) banking sub-domain applicability; and (vii) a brief weight justification.

Weights within each sub-score sum to 1.00; a sub-score’s value is the capped sum of the weights of the rules that fire. The overall *quality* score is the geometric mean of the five sub-score values, chosen so that weakness in any single dimension penalizes the overall verdict rather than being compensated by strength in another.

Groundedness (5 rules, weights sum to 1.00)

Rule 1 — claim_supported_by_context (weight 0.25, sub-score: Groundedness, tracks: 5/5) Description. Every factual claim in the rationale must be inferable from the provided retrieval context. Unsupported claims are flagged.

Provenance.

- *Academic.* Es et al. (2024), EACL 2024, §3 Faithfulness: “the answer $a_s(q)$ is faithful to the context $c(q)$ if the claims that are made in the answer can be inferred from the context.”
- *Academic, cross.* Honovich et al. (2022), NAACL 2022, §4: “large-scale NLI and question generation-and-answering-based approaches achieve strong and complementary results.”
- *Regulator.* National Institute of Standards and Technology (2024), NIST AI 600-1, p. 3 and §2.2 Confabulation: “The production of confidently stated but erroneous or false content (known colloquially as ‘hallucinations’ or ‘fabrications’) by which users may be misled or deceived.”
- *Regulator.* European Parliament and Council (2024), Art. 13(3)(b)(ii): “the level of accuracy, including its metrics, robustness and cybersecurity ... against which the high-risk AI system has been tested and validated.”
- *Industry, RAG benchmark.* Saad-Falcon et al. (2024), NAACL 2024, §3.1: *answer faithfulness* metric.
- *Industry, commercial.* Vectara HHEM-2.1-Open (Vectara, 2024) provides an open-weights NLI-style faithfulness classifier directly applicable to this rule.
- *Tier-1 bank.* UBS Group (2024): UBS public AI strategy materials explicitly cite the LLM tendency to “confidently ‘hallucinate’ ” as a deployment design input.

Cross-domain applicability. Universal — credit (rationale referencing borrower data), AML (rationale referencing transaction patterns), KYC (rationale referencing entity registries), fraud, sanctions, concentration, model risk.

Weight justification. Maximum weight in Groundedness. The rule is the operational core of LLM hallucination detection: every other Groundedness rule supports it. Five-track consensus combined with universal cross-domain applicability and direct mapping to EU AI Act Art. 13 makes it the load-bearing rule of the entire set.

Rule 2 — atomic_decomposition_per_claim (weight 0.20, Groundedness, 5/5) Description. Long rationales must be decomposable into atomic facts, each independently verifiable against the source context. Compound or holistic verdicts are insufficient.

Provenance.

- *Academic.* Min et al. (2023), EMNLP 2023, §3: “breaks a generation into a series of atomic facts and computes the percentage of atomic facts supported by a reliable knowledge source.”
- *Academic, cross.* Es et al. (2024), EACL 2024, §3 statement-level verification: “ $F = |V|/|S|$, where $|V|$ is the number of statements that were supported.”
- *Regulator.* European Parliament and Council (2024), Art. 13(3)(b)(iv): “the technical capabilities and characteristics of the high-risk AI system to provide information that is relevant to explain its output.”
- *Industry, RAG benchmark.* Friel et al. (2024), 2024, TRACe framework decomposition.
- *Tier-1 bank.* Morgan Stanley and OpenAI (2024): “Morgan Stanley met this challenge by implementing an evaluation (eval) framework to test every AI use case before deployment.”

Cross-domain applicability. Universal — particularly critical for credit decisioning (each underwriting factor independently verifiable) and AML (each transaction-pattern claim independently checkable).

Weight justification. Foundational methodology. Without atomic decomposition, holistic faithfulness scoring fails on rationales that mix supported and unsupported claims — the original motivation of Min et al. (2023).

Rule 3 — resists_unsupported_extensions (weight 0.20, Groundedness, 5/5) Description. The rationale does not introduce claims that go beyond what the retrieved context can support (no extrinsic hallucination).

Provenance.

- *Academic.* Ji et al. (2023), ACM Computing Surveys 55(12): intrinsic vs. extrinsic hallucination taxonomy.
- *Academic.* Li et al. (2023), EMNLP 2023: “content that conflicts with the source or cannot be verified by the factual knowledge.”
- *Industry, RAG benchmark.* Friel et al. (2024), TRACe adherence: “measures how faithfully the generated answer reflects the provided context.”
- *Regulator.* National Institute of Standards and Technology (2024): confabulation as named risk category.
- *Tier-1 bank.* Bank of America (2024): Bank of America’s “can do versus should do” design principle for customer-facing assistants, explicitly built on a controlled proprietary platform to limit extrinsic generation.

Cross-domain applicability. Universal. Particularly critical for sanctions screening (rationale must not infer matches beyond OFAC/EU list) and KYC (rationale must not assume facts beyond verified registries).

Weight justification. Extrinsic hallucinations are the most expensive failure mode in banking: they generate false positives in sanctions, false negatives in fraud, and adverse-action errors in credit.

Rule 4 — identifies_regulatory_or_policy_flag (weight 0.20, Groundedness, 4/5) Description. When applicable to the case, the rationale names a specific regulatory flag (AML typology, KYC element, sanctions list, insider, concentration limit) or internal policy clause rather than gesturing at generic “risk.”

Provenance.

- *Academic.* [Chen et al. \(2023\)](#), ACL 2023: “quantify the amount of new, label-relevant information in a rationale beyond the information already available in the input or the label.”
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 13(3)(b)(iv): explain-output capability.
- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §VI Documentation: “Adequate documentation helps to support effective model risk management. For example, documentation can help maximize the likelihood of continuity of operations, including supporting the tracking of recommendations, responses, and exceptions; it can also be used to more effectively help manage any model remediation efforts.”
- *Regulator.* [European Central Bank \(2024\)](#), §9.7.2 ¶61, cross-referencing [European Banking Authority \(2017\)](#) ¶58: “the ECB expects institutions to ensure the economic rationale of ML-based internal models and their risk drivers . . . institutions are expected to consult relevant experts from business areas with respect to the business rationale and risk contribution of the considered selection of risk drivers and rating criteria.”
- *Tier-1 bank.* [HSBC Holdings \(2024\)](#), Principle 3: “We identify the legitimate purpose of any use of data and AI.”
- *Financial NLP.* [Islam et al. \(2023\)](#): evidence-string protocol with page references back to source filings.

Cross-domain applicability. Highest-leverage cross-domain rule. AML rationales must name the typology; KYC must name the CDD element; sanctions must name the list; credit must name the adverse-action code (FCRA / ECOA); concentration must name the limit. The meta-pattern (concrete flag identified) is universal even though the specific flag set differs.

Weight justification. Banking-specific operationalization of REV. A rationale that says “risk detected” without naming the rule fails documentation requirements under SR 26-2 §VI and operational utility.

Rule 5 — `counterfactual_robustness_to_wrong_retrieval` (weight 0.15, Groundedness, 4/5) Description. When the retrieved context contains factually incorrect information, the rationale must not adopt or propagate it.

Provenance.

- *Academic.* [Chen et al. \(2024\)](#), AAAI 2024: “counterfactual robustness . . . scenarios where the model encounters contradictory information.”
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 15(4): “shall be as resilient as possible regarding errors, faults or inconsistencies . . . robustness . . . may be achieved through technical redundancy solutions.”
- *Cross-industry.* [National Institute of Standards and Technology \(2023\)](#), MEASURE function: robustness as core trustworthiness characteristic.
- *Cross-industry.* [International Organization for Standardization \(2023a\)](#): AI risk management standard with adversarial robustness.

Cross-domain applicability. Critical for KYC (stale/contradictory registries), sanctions (conflicting list versions), and model risk (superseded model documentation). Less critical for fraud (where evidence is usually transactional and recent).

Weight justification. Lower than R1–R4 because the failure mode is downstream of retrieval quality. A robust rule set assumes some retrieval noise is inevitable.

Completeness (3 rules, weights sum to 1.00)

Rule 6 — addresses_all_parts_of_question (weight 0.40, Completeness, 4/5) **Description.** When the governance question has multiple parts, the rationale must address each one; partial answers are flagged.

Provenance.

- *Academic.* Es et al. (2024), EACL 2024, §3 Answer Relevance: “penalises cases where the answer is incomplete or where it contains redundant information.”
- *Regulator.* European Parliament and Council (2024), Art. 13(2): “instructions for use ... include concise, complete, correct and clear information that is relevant, accessible and comprehensible to deployers.”
- *Regulator.* Bank of England, Prudential Regulation Authority (2023), Principle 3 Model development, implementation, and use: “Firms should have a robust model development process with standards for model design and implementation, model selection, and model performance measurement. Testing of data, model construct, assumptions, and model outcomes should be performed regularly in order to identify, monitor, record, and remediate model limitations and weaknesses.”
- *Industry, RAG benchmark.* Friel et al. (2024), TRACe completeness: “assesses whether the answer fully addresses the question.”
- *Cross-industry.* Monetary Authority of Singapore Veritas Consortium (2022), Transparency Methodology Doc 3C: completeness as part of assessment.

Cross-domain applicability. Universal — particularly for credit (must address all 5 Cs: Character, Capacity, Capital, Collateral, Conditions), KYC (all CDD elements), model risk (all SR 26-2 prongs).

Weight justification. Highest weight in Completeness. Partial rationales are a common LLM failure mode and they directly undermine regulator documentation requirements.

Rule 7 — covers_required_governance_dimensions

(weight 0.35, Completeness, 3/5) **Description.** The rationale references all the governance dimensions that the sub-domain requires (e.g., for credit: borrower, transaction, security, market; for AML: typology, customer profile, transaction pattern, jurisdiction).

Provenance.

- *Regulator.* European Banking Authority (2020), §4.3.3: “Understand the underlying models used, including their capabilities, assumptions and limitations, along with ensuring their traceability, auditability, and robustness and resilience.”
- *Regulator.* Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency (2026), §IV Model Development: conceptual soundness across dimensions, including data selection, methodology, and developmental testing.
- *Tier-1 bank.* HSBC Holdings (2024), Principle 5: “We test and monitor whether AI systems are working as they should both before being deployed and on an ongoing basis.”
- *Tier-1 bank.* Monetary Authority of Singapore (2018), Transparency: clear explanations of data inputs, mechanism, and consequences for affected subjects.
- *Financial NLP.* Islam et al. (2023): multi-dimensional question evaluation in enterprise financial QA.

Cross-domain applicability. The cross-domain backbone. The list of required dimensions is sub-domain-specific but the meta-rule (coverage check) is universal.

Weight justification. Operationalizes domain expertise; distinguishes a generic LLM answer from a banking-grade one.

Rule 8 — `information_integration_across_sources` (weight 0.25, Completeness, 3/5) Description. When the rationale draws on multiple retrieved documents, it integrates them coherently rather than concatenating or favoring one arbitrarily.

Provenance.

- *Academic.* [Chen et al. \(2024\)](#), AAAI 2024: “information integration . . . face challenges in combining information from multiple documents.”
- *Academic.* [Honovich et al. \(2022\)](#), NAACL 2022: multi-source NLI.
- *Tier-1 bank.* [Morgan Stanley and OpenAI \(2024\)](#): regression suite covers multi-document queries.

Cross-domain applicability. Concentration risk (aggregating exposures across entities), AML (combining multi-transaction patterns), credit (cross-bureau merging).

Weight justification. Specific failure mode — distinct from completeness but related.

Calibration (4 rules, weights sum to 1.00)

Rule 9 — `abstains_when_evidence_insufficient` (weight 0.35, Calibration, 5/5) Description. When the retrieved context does not contain enough information to answer the question, the rationale must explicitly state this rather than fabricate.

Provenance.

- *Academic.* [Es et al. \(2024\)](#), EACL 2024, §3 Context-Relevance prompt: “If no relevant sentences are found, or if you believe the question cannot be answered from the given context, return the phrase ‘Insufficient Information.’ ”
- *Academic.* [Manakul et al. \(2023\)](#), EMNLP 2023: divergence in samples as low-confidence flag.
- *Financial NLP.* [Islam et al. \(2023\)](#): refusal-rate analysis: “GPT-4-Turbo used with a retrieval system incorrectly answered or refused to answer 81% of questions.”
- *Financial NLP.* [Chen et al. \(2024\)](#), *negative rejection*: “when a model attempts to provide an answer despite lacking sufficient knowledge.”
- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §V Model Validation and Monitoring: “Sound validation identifies model limitations and errors and clarifies appropriate use and whether corrective actions may be warranted.”
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 13(3)(b)(iii): foreseeable-misuse disclosure.
- *Cross-industry.* [FinRegLab \(2023\)](#): calibration-uncertainty as governance criterion.
- *Tier-1 bank.* [UBS Group \(2024\)](#): explicit acknowledgment of “confidently wrong” failure mode.

Cross-domain applicability. Universal — but most critical for sanctions (false-positive cost) and AML (Suspicious Activity Report over-filing cost). A high-quality “I don’t know” beats a fabricated answer in regulated contexts.

Weight justification. Highest weight in Calibration. The abstain capability is the single most regulator-friendly feature an LLM-based system can have.

Rule 10 — expresses_uncertainty_explicitly (weight 0.30, Calibration, 5/5) Description. When the rationale is not certain of a claim, it uses explicit hedging language (*may, suggests, appears to indicate*) rather than confident assertion.

Provenance.

- *Academic.* Lin et al. (2022), ACL 2022: “models must avoid generating false answers learned from imitating human texts.”
- *Academic.* Liu et al. (2023), EMNLP 2023: probability-weighted scoring captures graded confidence.
- *Academic, linguistic.* Hyland (1998): foundational taxonomy of hedging.
- *Regulator.* Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency (2026), §IV Model Use: “Effective model use depends on a clear understanding of a model’s limitations.”
- *Regulator.* National Institute of Standards and Technology (2024): confidence calibration in the GenAI Profile.
- *Cross-industry.* FinRegLab (2023): calibration-uncertainty as governance criterion.
- *Tier-1 bank.* UBS Group (2024); Monetary Authority of Singapore (2024): explicit acknowledgement of overconfidence as named GenAI risk dimension.
- *Financial NLP.* Askill et al. (2021) (BIG-bench HHH honesty subtask): the honesty/accuracy category is one of the four pragmatically broken-down alignment categories that the task evaluates.

Cross-domain applicability. Universal. Especially impactful for credit (adverse-action language must reflect confidence level) and AML (analyst-facing alerts must distinguish “likely typology” from “confirmed match”).

Weight justification. Strong cross-track consensus plus clear regulatory framing.

Rule 11 — labels_uncertainty_with_confidence_score (weight 0.20, Calibration, 4/5) Description. Where applicable, the rationale carries a numeric confidence score (probability, likelihood, percentile) that is calibrated to actual outcome.

Provenance.

- *Academic.* Liu et al. (2023): probability-weighted-judge-score methodology.
- *Regulator.* European Parliament and Council (2024), Art. 13(3)(b)(ii): “level of accuracy, including its metrics.”
- *Regulator.* National Institute of Standards and Technology (2023), MEASURE 2.3/2.5: calibration metrics.
- *Cross-industry.* KPMG International (2023): system-cards approach with confidence disclosure.
- *Industry, commercial.* Vectara HHEM (Vectara, 2024) produces a scored faithfulness probability.

Cross-domain applicability. Strongest in credit (PD, LGD already scored) and fraud (fraud score is native). Less natural in qualitative KYC narrative work.

Weight justification. Lower weight; many banking sub-domains operate in qualitative narrative mode where numeric calibration is unnatural.

Rule 12 — internal_self_consistency_check (weight 0.15, Calibration, 3/5) Description. When the system is re-queried with the same prompt at non-zero temperature, the rationale should not contradict itself across samples.

Provenance.

- *Academic.* Manakul et al. (2023), EMNLP 2023: “if an LLM has knowledge of a given concept, sampled responses are likely to be similar and contain consistent facts. However, for hallucinated facts, stochastically sampled responses are likely to diverge and contradict one another.”
- *Tier-1 bank.* Morgan Stanley and OpenAI (2024): “Daily testing with a regression suite of sample questions identified potential weaknesses.”
- *Financial NLP.* Su et al. (2024): cross-source consistency.

Cross-domain applicability. Universal but operationally expensive (requires multi-sampling). Most useful for high-stakes decisions (concentration limit breaches, large credit exposures).

Weight justification. Lowest weight in Calibration; the cost-of-application is high relative to typical banking inference budgets.

Traceability (5 rules, weights sum to 1.00)

Rule 13 — cites_specific_source_span (weight 0.25, Traceability, 5/5) Description. Each non-trivial claim points to a specific source span (document ID, paragraph, table cell) rather than gesturing at “the documentation” or “the policy.”

Provenance.

- *Academic.* Camburu et al. (2018), NeurIPS 2018: “obtaining full sentence justifications of a model’s decisions.”
- *Academic.* Honovich et al. (2022): QAG-based span verification.
- *Regulator.* European Parliament and Council (2024), Art. 13(3)(b)(iv); Art. 12 Record-Keeping.
- *Regulator.* Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency (2026), §VI Documentation.
- *Cross-industry.* 12 of 14 frameworks surveyed in Track 4 include traceability or logging as an explicit pillar.
- *Tier-1 bank.* Monetary Authority of Singapore Veritas Consortium (2022), Transparency Methodology Doc 3C: internal versus external transparency operationalization.
- *Tier-1 bank.* HSBC Holdings (2024), Principle 1: human-justification standard.
- *Financial NLP.* Islam et al. (2023): each entry includes “an evidence string (which contains the information needed to verify that the answer to the question is correct) and a page number from the relevant document.”
- *Industry, RAG benchmark.* Friel et al. (2024): TRACe utilization.

Cross-domain applicability. Universal — the foundational audit-trail requirement that every regulator and framework demands.

Weight justification. Maximum weight in Traceability. Without a traceable citation, audit defense fails.

Rule 14 — provides_natural_language_rationale (weight 0.20, Traceability, 5/5) Description. The system produces a full natural-language justification for each decision, not just a label or score.

Provenance.

- *Academic.* Camburu et al. (2018): full sentence justifications.

- *Academic.* [Chen et al. \(2023\)](#): natural-language rationale informativeness.
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 13(3)(b)(iv): explain-output capability.
- *Regulator.* [Bank of England, Prudential Regulation Authority \(2023\)](#), Principle 3: “Firms should have a robust model development process with standards for model design and implementation, model selection, and model performance measurement.”
- *Regulator.* [Bank of England and Financial Conduct Authority \(2022\)](#), cross-cutting principles: “make sure that AI is appropriately transparent and explainable.”
- *Regulator.* [European Central Bank \(2024\)](#), §9.7.2 ¶60: “The ECB expects institutions to motivate the complexity of their internal models and avoid unnecessarily complex modelling approaches by taking into account any trade-off between, on the one hand, the complexity of the modelling techniques and, on the other, the performance increase of the ML-based internal models.”
- *Cross-industry.* 13 of 14 frameworks include explainability as a pillar.
- *Tier-1 bank.* [Citi Ventures \(2024\)](#): explainability/observability/transparency gate.
- *Financial NLP.* [Chen et al. \(2021\)](#): program-accuracy metric over reasoning steps.

Cross-domain applicability. Universal. Particularly critical for credit (FCRA/ECOA adverse-action notice requires natural-language explanation) and AML (Suspicious Activity Report narrative).

Weight justification. Strong cross-track consensus but lower than R13: existence of a natural-language rationale is necessary but not sufficient for traceability.

Rule 15 — `provides_falsifiable_actionable_claim` (weight 0.20, Traceability, 3/5)

Description. The rationale’s claims are specific enough to be testable or refutable; vague generalities are flagged.

Provenance.

- *Academic.* [Chen et al. \(2023\)](#): “quantify the amount of new, label-relevant information in a rationale beyond the information already available in the input or the label.”
- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §V Conceptual Soundness: “Validating conceptual soundness involves assessing and documenting model design (including key modeling choices, assumptions, qualitative judgments, and data selection), construction, and developmental testing.”
- *Regulator.* [European Banking Authority \(2017\)](#), ¶58 (as cross-referenced by [European Central Bank \(2024\)](#) ¶61): business-rationale consultation requirement.
- *Financial NLP.* [Islam et al. \(2023\)](#): evidence-string grounding with page references.

Cross-domain applicability. Universal cross-domain test of rationale quality. Particularly important for credit (vague adverse-action language fails ECOA) and AML (vague SAR narratives are rejected by FinCEN).

Weight justification. Operationalizes the concrete-versus-vague axis of rationale quality.

Rule 16 — `includes_numeric_value_or_metric` (weight 0.15, Traceability, 3/5)

Description. When the case involves quantifiable risk (amount, threshold, score, ratio), the rationale includes the relevant numeric value(s).

Provenance.

- *Financial NLP.* [Chen et al. \(2021\)](#), EMNLP 2021: execution-accuracy metric for numerical reasoning.

- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 13(3)(b)(ii): accuracy metrics disclosed.
- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §V Outcomes Analysis: “Outcomes analysis compares model outputs to corresponding real-world outcomes to assess model performance relative to model objectives and business use.”
- *Tier-1 bank.* [Morgan Stanley and OpenAI \(2024\)](#): eval framework includes numerical regression.

Cross-domain applicability. Strongest in credit (DTI, LTV, PD, LGD), concentration (limit calculations), market risk. Weaker in qualitative AML narrative work.

Weight justification. Sub-domain-dependent; should fire only when numeric values are applicable.

Rule 17 — `logs_decision_for_audit` (weight 0.20, Traceability, 3/5) Description. The rationale, with timestamp and model version, is persisted to an immutable audit log.

Provenance.

- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 12 Record-Keeping (cross-referenced by Art. 13(3)(f)): “description of the mechanisms . . . that allows deployers to properly collect, store and interpret the logs in accordance with Article 12.”
- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §VI Documentation: “Adequate documentation helps to support effective model risk management.”
- *Cross-industry.* [International Organization for Standardization \(2023b\)](#), §8.2 AI lifecycle management.
- *Tier-1 bank.* [JPMorgan Chase \(2023\)](#): model risk governance applied to LLMs.

Cross-domain applicability. Universal — required for every sub-domain. The GroundLens audit module implements SHA-256 hash-chain logging matching this requirement.

Weight justification. Critical for audit defense; already implemented in the companion library’s audit module.

Robustness (3 rules, weights sum to 1.00)

Rule 18 — `references_independent_validation` (weight 0.40, Robustness, 3/5) Description. The rationale or its production pipeline references independent validation by a party other than the model developer (effective challenge).

Provenance.

- *Regulator.* [Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency \(2026\)](#), §III Effective Challenge: “Effective challenge is performed by individuals with the appropriate expertise to conduct a critical and objective challenge, sufficient independence to maintain objectivity, as well as the organizational standing and influence to effect any change.”
- *Regulator.* [Bank of England, Prudential Regulation Authority \(2023\)](#), Principle 4 Independent model validation: “Firms should have a validation process that provides ongoing, independent, and effective challenge to model development and use.”
- *Regulator.* [European Central Bank \(2024\)](#), §9.3 ¶43(a): “Internal validation should effectively challenge the modelling decisions taken regarding ML-based internal models, assessing whether their complexity is justified, and evaluating the stability and robustness of the outcomes.”

- *Regulator.* [European Banking Authority \(2020\)](#): lifecycle governance and validation requirements for credit models.
- *Cross-industry.* [International Organization for Standardization \(2023b\)](#): governance requirements.
- *Tier-1 bank.* [JPMorgan Chase \(2023\)](#): “Explainable AI, Responsible AI and Ethical AI ... are no longer being seen as optional but required.”

Cross-domain applicability. Universal — every regulator names independent challenge as a P0 requirement.

Weight justification. Highest weight in Robustness; the strongest cross-regulator consensus of any rule in the corpus.

Rule 19 — resists_prompt_injection_or_distractor (weight 0.35, Robustness, 3/5)

Description. The rationale quality does not degrade when retrieval surfaces irrelevant or adversarial content alongside relevant context.

Provenance.

- *Academic.* [Chen et al. \(2024\)](#): noise-robustness: “addresses situations where the retrieved information includes irrelevant or inaccurate content.”
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 15(1)+(4): robustness requirement and resilience to errors.
- *Cross-industry.* [International Organization for Standardization \(2023a\)](#): security and adversarial robustness.
- *Cross-industry.* [Monetary Authority of Singapore \(2024\)](#): prompt injection named as GenAI risk dimension.

Cross-domain applicability. Especially critical for AML (alert-rich case files) and KYC (large customer dossiers).

Weight justification. Specific failure mode in RAG-based banking systems.

Rule 20 — handles_cross_source_conflict (weight 0.25, Robustness, 3/5)

Description. When retrieved sources disagree, the rationale identifies the conflict explicitly and reasons about reliability rather than silently selecting one source.

Provenance.

- *Academic.* [Su et al. \(2024\)](#): “LLMs may struggle to identify the more reliable evidence ... when facing such cross-source conflicts.”
- *Academic.* [Chen et al. \(2024\)](#): counterfactual-robustness.
- *Regulator.* [European Parliament and Council \(2024\)](#), Art. 15(4): resilience to inconsistencies.

Cross-domain applicability. KYC (registry versus customer attestation), sanctions (OFAC versus EU list mismatch), credit (bureau disagreement).

Weight justification. Real failure mode but currently rare in well-curated banking corpora.

4.3 Honorable mentions: 30 candidates retained for future expansion

Thirty further candidates received cross-track support score $S(r) < 3$ or were subsumed semantically by a top-20 rule but retain value as the source pool for sub-domain-specific rule sets (*e.g.*, an AML-specific extension that imports the top-20 plus AML-specific rules). Table 2 summarises them.

Table 2: Thirty honorable mentions retained for sub-domain expansion or future Groundedness-focused rulesets.

Candidate ID	Description	Primary source(s)
<i>Academic (10)</i>		
resistance_to_imitative_falsehoods	Resistance on human misconceptions	TruthfulQA (Lin et al., 2022)
context_relevance_focus	Retrieved context contains only relevant content	RAGAs (Es et al., 2024) (system-level)
zero_resource_blackbox_hallucination	Hallucination detection without external KB	SelfCheckGPT (Manakul et al., 2023) (methodology)
cot_form_filling_evaluation	LLM-as-judge with CoT + form-filling	G-Eval (Liu et al., 2023) (meta-eval)
probability_weighted_judge_score	Take score probability weighted judge score	G-Eval (Liu et al., 2023)
llm_judge_self_bias_warning	Using same model as generator + judge	G-Eval (Liu et al., 2023) §4
external_knowledge_or_reasoning_knowledge	Reasoning and knowledge improves detection	HaluEval (Li et al., 2023)
intrinsic_vs_extrinsic_hallucination	Hallucination taxonomy	Ji survey (Ji et al., 2023)
example_level_meta_eval	Per-example accuracy of consistency metrics	TRUE (Honovich et al., 2022)
new_information_in_rationale	Verifiable measure of rationale content	REV (Chen et al., 2023) (partly captured in R15)
<i>Tier-1 bank (5)</i>		
human_justification_test	Decision must be justifiable as if human-made	HSBC (HSBC Holdings, 2024), P1
purpose_legitimacy_documented	Intended AI use must have documented purpose	HSBC (HSBC Holdings, 2024), P3
eval_framework_pre_deployment	Every use case has pre-deployment eval	Morgan Stanley (Morgan Stanley and OpenAI, 2024) (system-level)
daily_regression_quality	Daily regression suite against fixed questions	Morgan Stanley (Morgan Stanley and OpenAI, 2024) (system-level)
controlled_proprietary_platform_for_customer_facing	Platform for Customer-facing	Bank of America (Bank of America, 2024) (architecture)
<i>Regulator (10)</i>		
srl_outcomes_analysis_presentation	Presenting / benchmarking evidence cited	SR 26-2 (Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026) §V Outcomes Analysis

Candidate ID	Description	Primary source(s)
srl_ongoing_monitoring_triggering	Monitoring triggers for accelerated review	SR 26-2 (Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026) §V Ongoing Model Monitoring
srl_conceptual_soundness_check	Quality of design and methodology assessed	SR 26-2 (Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, 2026) §V Conceptual Soundness
eu_ai_act_15_resilient_Technicals	Technical redundancy for resilience	EU AI Act (European Parliament and Council, 2024) Art. 15(4)
eu_ai_act_15_consistent_Performance	Performance accuracy across lifecycle	EU AI Act (European Parliament and Council, 2024) Art. 15(1)
eu_ai_act_13_subgroup_Performance	Performance disclosed by subgroup	EU AI Act (European Parliament and Council, 2024) Art. 13(3)(b)(v) (fairness)
pra_ss123_principle1_id	Model to classify risk tiering	PRA SS1/23 (Bank of England, Prudential Regulation Authority, 2023) Principle 1
nist_ai_rmf_govern_function	GOVERN function activities	NIST AI RMF (National Institute of Standards and Technology, 2023)
fca_dp522_explainable_principle	“Appropriately transparent and explainable” principle	FCA DP5/22 (Bank of England and Financial Conduct Authority, 2022)
ecb_im_ml_explainability_justifies_complexity	ML justifies complexity motivated	ECB Guide (European Central Bank, 2024) §9.7.2
<i>Cross-industry and Financial NLP (5)</i>		
impact_assessment	Pre-deployment impact assessment	ISO/IEC 42001 (International Organization for Standardization, 2023b)
third_party_supply_chain_foundation	Foundation model vendor risk assessed	NIST AI 600-1 (National Institute of Standards and Technology, 2024)
fairness_subgroup_evaluation	Subgroup fairness metrics	14/14 frameworks (governance)
privacy_minimization	Data minimization for personal data	14/14 frameworks (governance)

Candidate ID	Description	Primary source(s)
expert_economic_rationality	Any rationale must explain absence of economic rationality	AML literature (Anonymous authors, 2025) (domain-specific)

5 Discussion

5.1 Why these five categories, and not others

A natural objection to the five-category skeleton is that other taxonomies are conceivable — for instance, the FEAT four-pillar structure (fairness / ethics / accountability / transparency), or the NIST AI RMF trustworthiness characteristics (*valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, fair with harmful bias managed*).

We chose the five-category skeleton on three grounds. First, it emerges from the data: the categories were not pre-specified before the parallel research design and they survived the cross-track clustering with substantial within-category internal coherence (all rules in Groundedness operationalize a claim-to-source linkage; all rules in Calibration operationalize uncertainty handling, etc.). Second, the categories operate at the right level of abstraction for a rule-based evaluation library: they are coarse enough to admit several rules each but fine enough that the geometric-mean aggregation formula is meaningful. Third, the categories align closely with regulator vocabulary, reducing friction in supervisory dialogue: *Groundedness* maps to EU AI Act Art. 13 and NIST 600-1 Confabulation; *Calibration* maps to SR 26-2 §V validation language and EU AI Act Art. 13(3)(b)(ii); *Traceability* maps to EU AI Act Art. 12 and SR 26-2 §VI Documentation; *Robustness* maps to EU AI Act Art. 15 and SR 26-2 §III Effective Challenge; *Completeness* maps to EU AI Act Art. 13(2).

The FEAT four-pillar structure operates at the governance level (fairness, ethics, accountability, transparency) rather than the rule-evaluable level: “fairness” is not directly testable on a single rationale in the way “claim supported by context” is. The NIST trustworthiness characteristics are well-suited to system-level governance but cover a wider field than rule-evaluable rationale quality (*privacy-enhanced, safe, fair with harmful bias managed* are properties of training-data and deployment governance, not properties of an individual rationale evaluated by a rule).

5.2 Defendability analysis

A rule set’s defendability to a regulator depends on two distinct properties: (i) the diversity and independence of its source corpus, and (ii) the transparency of its construction methodology. The present rule set scores well on both axes.

Source diversity. Of the 20 rules, all 20 cite at least one academic source, all 20 cite at least one regulator or cross-industry framework source, and 13 cite at least one tier-1 bank source. The ideal pattern (one academic, one industry, one regulator) is satisfied by 13 of 20; the remaining 7 satisfy the weaker but still-acceptable pattern of one academic plus two regulator-or-cross-industry sources. No rule in the top-20 relies on a single source.

Construction methodology. The five-track parallel research design with source verification is documented in §3 and is reproducible. The deduplication and merging decisions are explicit per-rule. The skeleton derivation is empirical rather than ex-ante. These properties together meet the standard articulated in SR 26-2 §III: “Effective challenge is performed by individuals with the appropriate expertise to conduct a critical and objective challenge, sufficient independence

to maintain objectivity, as well as the organizational standing and influence to effect any change.” Independent challenge here is operationalized at the rule-source level rather than the system-development level.

Regulator vocabulary alignment. The five sub-score categories (Groundedness, Completeness, Calibration, Traceability, Robustness) map directly to specific clauses of the EU AI Act, the NIST AI RMF, SR 26-2, and PRA SS1/23. This mapping reduces friction in supervisory dialogue: a regulator asking “how does your tool address EU AI Act Art. 13 transparency?” can be answered “via the Traceability sub-score, which aggregates rules R13–R17, each citing Art. 13 or analogous SR 26-2 clauses.”

6 Limitations

6.1 Tier-1 bank disclosure asymmetry

Public AI governance disclosure is highly uneven across tier-1 banks. HSBC, MAS (via FEAT and Veritas), and Morgan Stanley (via the OpenAI collaboration case study) provide substantial public material. JPMorgan, UBS, Citi, and Bank of America publish governance positions via press statements and interviews but no formal framework documents. BlackRock, Goldman Sachs, Deutsche Bank, Barclays, and Santander were not found to publish detailed AI rationale evaluation frameworks publicly. This asymmetry means that Track 2 evidence is concentrated in 5–7 institutions, not the full 12 surveyed. Cross-track support scores remain robust because Track 2 is one of five independent tracks, but a deployer evaluating an AI governance rule against a specific institution’s public framework should not assume coverage across the tier-1 banking sector.

6.2 Cross-domain generality versus sub-domain depth

The top-20 rule set is intentionally cross-domain. This means it sacrifices depth in any single sub-domain: AML-specific rules (typology recognition, economic-rationality narrative quality), credit-specific rules (FCRA/ECOA adverse-action reason-code stability), and KYC-specific rules (UBO chain reconciliation) are intentionally placed in honorable mentions rather than the top-20. Deployers who serve a single sub-domain should consider extending the top-20 with relevant sub-domain rules from §4.3.

6.3 Temporal currency

The corpus reflects sources accessible as of June 2026. Banking regulator publications are updated on multi-year cycles: the present rule set should be re-evaluated against new publications at least annually. SR 26-2 itself supersedes the prior fifteen years of US model-risk guidance — a precedent that suggests future updates may materially shift specific verbatim citations even where the underlying principles remain stable. Similarly, the EU AI Act delegated acts under Article 15 are being developed by the European Commission via metrology and standards bodies, and will likely surface additional canonical criteria.

6.4 Single-rater verification

The deduplication and merging decisions in §3.3, the skeleton clustering in §3.4, and the weight assignments in §4.2 were performed by a single rater (the author). Reproducibility under inter-rater verification is a desirable but not-yet-performed validation step.

7 Conclusion and Future Work

We have presented a methodology, and its execution, for constructing a rule set for LLM rationale evaluation in banking governance whose provenance is triangulated across five independent research tracks. The resulting top-20 rule set, organized into five empirically-emergent sub-score categories, is designed to be defensible in front of EU and US banking supervisors and Big Four model-risk auditors. Each rule is documented with at least three independent provenance citations following the ideal pattern of one academic, one industry, and one regulator source. The companion implementation is shipped in the open-source GroundLens library as the canonical `groundlens_banking_rules` reference.

Immediate follow-up. Two follow-up actions are required before the rule set can be deployed in client-facing settings: (i) Python implementation of the 20 rules as composable `ChecklistRule` instances in the GroundLens library, with each rule carrying a `citation: str` field populated from this paper; (ii) publication of the rule set under a versioned release with a CHANGELOG entry documenting the breaking change from the prior baseline ruleset.

Medium-term follow-up. Two extensions are natural. First, sub-domain-specific rule sets (AML extension, credit-decisioning extension, KYC extension, sanctions extension) can be constructed by composing the top-20 with selected honorable mentions and additional domain-specific rules. Second, the multi-source provenance framework presented here generalizes beyond banking governance: it could be applied to LLM rationale evaluation in healthcare (clinical decision support), legal (contract review), and insurance underwriting, each with its own regulator and tier-1 industry track.

Open question. The methodology assumes that cross-track independence is the right axis for defensibility. An alternative formulation would weight tracks differently (*e.g.*, regulator citations weighted more heavily than academic citations). We have not explored this systematically and leave it open as future work; empirically, the current uniform-weight formulation produces a rule set whose category structure aligns with regulator vocabulary, which suggests the formulation is at least not pathological.

References

- Anonymous authors. Generative AI for AML suspicious activity report narratives: Evaluation criteria. arXiv:2512.06240, 2025. URL <https://arxiv.org/abs/2512.06240>. Author attribution requires verification at publication time.
- Amanda Askill, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. A general language assistant as a laboratory for alignment. arXiv:2112.00861. BIG-bench HHH alignment task, 2021. URL <https://arxiv.org/abs/2112.00861>.
- Bank for International Settlements, Financial Stability Institute. Regulating AI in the financial sector: recent developments and main challenges. FSI Insights No 63. Technical report, Bank for International Settlements, 2024. URL <https://www.bis.org/fsi/publ/insights63.pdf>.
- Bank of America. Bank of america’s AI approach: “productive paranoia”. American Banker (secondary coverage) and AIX case study, 2024. URL <https://www.americanbanker.com/news/bank-of-americas-ai-approach-productive-paranoia>.

- Bank of England and Financial Conduct Authority. Artificial intelligence and machine learning. discussion paper DP5/22. Technical report, Bank of England and FCA, 2022. URL <https://www.bankofengland.co.uk/prudential-regulation/publication/2022/october/artificial-intelligence>.
- Bank of England, Prudential Regulation Authority. Model risk management principles for banks. supervisory statement SS1/23. Technical report, Bank of England, 2023. URL <https://www.bankofengland.co.uk/prudential-regulation/publication/2023/may/model-risk-management-principles-for-banks-ss>.
- Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency. Revised supervisory guidance on model risk management. SR letter 26-2, Federal Reserve, FDIC, and OCC, April 2026. URL <https://www.federalreserve.gov/supervisionreg/srletters/SR2602.pdf>. Supersedes SR Letter 11-7 (April 2011) and SR Letter 21-8 (April 2021).
- Oana-Maria Camburu, Tim Rocktäschel, Thomas Lukasiewicz, and Phil Blunsom. e-SNLI: Natural language inference with natural language explanations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. URL <https://arxiv.org/abs/1812.01193>.
- Hanjie Chen, Faeze Brahman, Xiang Ren, Yangfeng Ji, Yejin Choi, and Swabha Swayamdipta. REV: Information-theoretic evaluation of free-text rationales. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023. URL <https://aclanthology.org/2023.acl-long.112/>.
- Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. Benchmarking large language models in retrieval-augmented generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024. URL <https://arxiv.org/abs/2309.01431>.
- Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. FinQA: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021. URL <https://arxiv.org/abs/2109.00122>.
- Citi Ventures. Citi gen AI summit 2024: Takeaways. Citi Ventures Perspectives, 2024. URL https://www.citi.com/ventures/perspectives/news/citi-gen-ai-summit-2024_gen-ai-citi.html.
- Rodrigo De La Chica and Jordi Martí-González. Auditable hallucination triage with mechanical enforcement: A rule-based framework for LLM rationale evaluation. arXiv preprint arXiv:2605.14744, 2026. URL <https://arxiv.org/abs/2605.14744>. Preprint, not peer-reviewed at time of writing.
- Deloitte. Trustworthy AI framework. Technical report, Deloitte, 2024. URL <https://www2.deloitte.com/us/en/pages/deloitte-analytics/solutions/ethics-of-ai-framework.html>.
- Ernst & Young. Responsible AI principles and trusted AI platform. Technical report, EY, 2024. URL https://www.ey.com/en_gl/consulting/how-do-you-teach-ai-the-value-of-trust.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAs: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL): System Demonstrations*, 2024. URL <https://arxiv.org/abs/2309.15217>.

- European Banking Authority. Guidelines on PD estimation, LGD estimation and the treatment of defaulted exposures. EBA/GL/2017/16. Technical report, European Banking Authority, 2017. URL <https://www.eba.europa.eu/regulation-and-policy/credit-risk/guidelines-on-pd-estimation-lgd-estimation-and-the-treatment-of-defaulted-exposures>.
- European Banking Authority. Guidelines on loan origination and monitoring. EBA/GL/2020/06. Technical report, European Banking Authority, 2020. URL <https://www.eba.europa.eu/regulation-and-policy/credit-risk/guidelines-on-loan-origination-and-monitoring>.
- European Central Bank. Guide to internal models. Technical report, European Central Bank Banking Supervision, 2024. URL https://www.bankingsupervision.europa.eu/ecb/pub/pdf/ssm_supervisory_guides202402_internalmodels.en.pdf. Revised version July 2025.
- European Parliament and Council. Regulation (EU) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence and amending certain union legislative acts (EU AI Act). Technical report, Official Journal of the European Union, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>. Full text and per-article browsing also at <https://artificialintelligenceact.eu/>.
- Richárd Farkas, Veronika Vincze, György Móra, János Csirik, and György Szarvas. The CoNLL-2010 shared task: Learning to detect hedges and their scope in natural language text. In *Proceedings of the Fourteenth Conference on Computational Natural Language Learning (CoNLL) – Shared Task*, 2010. URL <https://aclanthology.org/W10-3001/>.
- FinRegLab. Explainability and fairness in machine learning for credit underwriting: Empirical insights and policy analysis. Technical report, FinRegLab and Stanford Digital Economy Lab, 2023. URL <https://finreglab.org/research/explainability-and-fairness-in-machine-learning-for-credit-underwriting/>.
- Robert Friel, Atindriyo Belyi, and Soumyadip Sanyal. RAGBench: Explainable benchmark for retrieval-augmented generation systems. arXiv:2407.11005, 2024. URL <https://arxiv.org/abs/2407.11005>.
- H. Paul Grice. Logic and conversation. In Peter Cole and Jerry L. Morgan, editors, *Syntax and Semantics, Vol. 3: Speech Acts*, pages 41–58. Academic Press, New York, 1975.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. TRUE: Re-evaluating factual consistency evaluation. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, 2022. URL <https://arxiv.org/abs/2204.04991>.
- HSBC Holdings. Principles for the ethical use of data and AI. Technical report, HSBC, 2024. URL <https://www.hsbc.com/-/files/hsbc/our-approach/risk-and-responsibility/pdfs/240715-hsbc-principles-for-the-ethical-use-of-data-and-ai.pdf>.
- Ken Hyland. *Hedging in Scientific Research Articles*. John Benjamins, Amsterdam, 1998.
- Infocomm Media Development Authority (Singapore). AI Verify: Testing framework and toolkit for trustworthy AI, 2024. URL <https://aiverifyfoundation.sg/>.
- International Organization for Standardization. ISO/IEC 23894:2023 information technology — artificial intelligence — guidance on risk management. Technical report, ISO/IEC, 2023a. URL <https://www.iso.org/standard/77304.html>.

- International Organization for Standardization. ISO/IEC 42001:2023 information technology — artificial intelligence — management system. Technical report, ISO/IEC, 2023b. URL <https://www.iso.org/standard/42001>.
- Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. FinanceBench: A new benchmark for financial question answering. arXiv:2311.11944, 2023. URL <https://arxiv.org/abs/2311.11944>.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023. doi: 10.1145/3571730. URL <https://dl.acm.org/doi/10.1145/3571730>.
- JPMorgan Chase. AI and model risk governance. Corporate news article, 29 May 2023, 2023. URL <https://www.jpmorganchase.com/about/technology/news/ai-and-model-risk-governance>.
- KPMG International. The trusted AI approach. Technical report, KPMG, 2023. URL <https://kpmg.com/kpmg-us/content/dam/kpmg/pdf/2023/kpmg-trusted-ai-approach.pdf>.
- Maxim Krivorotov and Brandon Richey. Explainable AI for credit decisioning: Stability constraints from adverse-action requirements. SSRN 4133915, 2022. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4133915.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. URL <https://arxiv.org/abs/2305.11747>.
- Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. URL <https://arxiv.org/abs/2109.07958>.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. URL <https://arxiv.org/abs/2303.16634>.
- Annie Louis and Ani Nenkova. Automatic identification of general and specific sentences by leveraging discourse annotations. In *Proceedings of the 5th International Joint Conference on Natural Language Processing (IJCNLP)*, 2011. URL <https://aclanthology.org/I11-1067/>.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. URL <https://arxiv.org/abs/2303.08896>.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. URL <https://arxiv.org/abs/2305.14251>.
- Monetary Authority of Singapore. Principles to promote fairness, ethics, accountability and transparency (FEAT) in the use of AI and data analytics in singapore’s financial sector. Technical report, MAS, 2018. URL <https://www.mas.gov.sg/publications/monographs-or-information-paper/2018/feat>.

Monetary Authority of Singapore. Project MindForge: Emerging risks and opportunities of generative AI for banks. Technical report, MAS, 2024. URL <https://www.mas.gov.sg/-/media/mas-media-library/schemes-and-initiatives/ftig/project-mindforge/emerging-risks-and-opportunities-of-generative-ai-for-banks.pdf>.

Monetary Authority of Singapore Veritas Consortium. FEAT principles assessment methodology. documents 1, 2, 3a, 3b, 3c, 3d, 4. Technical report, MAS, 2022. URL <https://www.mas.gov.sg/news/media-releases/2022/mas-led-industry-consortium-publishes-assessment-methodologies-for-responsible-use-of-ai>.

Morgan Stanley and OpenAI. Morgan stanley uses AI evals to shape the future of financial services. OpenAI case study, 2024. URL <https://openai.com/index/morgan-stanley/>.

National Institute of Standards and Technology. Artificial intelligence risk management framework (AI RMF 1.0). NIST AI 100-1. Technical report, U.S. Department of Commerce, 2023. URL <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>.

National Institute of Standards and Technology. Artificial intelligence risk management framework: Generative artificial intelligence profile. NIST AI 600-1. Technical report, U.S. Department of Commerce, 2024. URL <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>.

Office of the Comptroller of the Currency. Model risk management. comptroller’s handbook. Technical report, OCC, 2021. URL <https://www.occ.gov/publications-and-resources/publications/comptrollers-handbook/files/model-risk-management/index-model-risk-management.html>.

Organisation for Economic Co-operation and Development. Recommendation of the council on artificial intelligence. OECD/LEGAL/0449. Technical report, OECD, 2019. URL <https://oecd.ai/en/dashboards/ai-principles/P5>. Updated 2024.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024. URL <https://arxiv.org/abs/2311.09476>.

Zhaochen Su, Jun Zhang, Xiaoye Qu, Yanshu Cheng, Hao Zhang, Jihui Chen, Bin Lin, and Yu Cheng. ConflictBank: A benchmark for evaluating the influence of knowledge conflicts in LLMs. arXiv:2408.12076, 2024. URL <https://arxiv.org/abs/2408.12076>.

TruLens Contributors. RAG Triad: Context relevance, groundedness, answer relevance. Open-source documentation, 2024. URL https://www.trulens.org/getting_started/core_concepts/rag_triad/.

UBS Group. Innovation and AI: Group strategy materials. Corporate website, 2024. URL <https://www.ubs.com/global/en/our-firm/what-we-do/technology/innovation-and-ai.html>.

Vectara. HHEM-2.1-Open: Hallucination evaluation model. Hugging Face model card, 2024. URL https://huggingface.co/vectara/hallucination_evaluation_model.

Veronika Vincze, György Szarvas, Richárd Farkas, György Móra, and János Csirik. The BioScope corpus: biomedical texts annotated for uncertainty, negation and their scopes. *BMC Bioinformatics*, 9(Suppl 11):S9, 2008. doi: 10.1186/1471-2105-9-S11-S9. URL <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/1471-2105-9-S11-S9>.